Ishrath Ahamed

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Experience

Software Engineer Jun. 2024 – Present
Cloud Solutions International (Pvt) Ltd *Sri Lanka*

- Designed and developed scalable, interoperable EHR microservices using Python (Django) and Java (Spring Boot), including IVF workflows
- Developed a production-grade agentic chatbot assistant for doctors on the EHR platform, powered by MCP servers, RAG, and Neo4j knowledge graphs
- Built a production-grade, real-time voice AI agent for an Android app over a low-latency WebRTC streaming pipeline
- Implemented a Cerbos policy manager service in Go for resource permissions and improved security service performance
- Established API Gateway policy baselines (CORS, rate limits, security headers, public endpoint exception lists)
- Built internal platforms to track microservice release dependencies and automated CI/CD pipelines (Jenkins, Liquibase, ArgoCD), cutting release time from 1 hour to under 10 minutes
- Led QA environment setup and managed Kubernetes-based deployments and containerized infrastructure
- Tech Stack: Go, Python, Django, Java, Spring Boot, Cerbos, Angular, Flutter, WebRTC, Kubernetes, Docker, Jenkins, Liquibase, ArgoCD, Neo4j

Visiting Research Fellow Jan. 2023 – Jun. 2023
Singapore University of Technology and Design *Singapore*

- Developed 3D CNN-based video classification models to detect illegal activities in residential surveillance
- Implemented face-recognition pipelines using deep feature embeddings for secure access control
- Designed lightweight people-counting and object-tracking algorithms optimized for edge deployment
- Tech Stack: Python, TensorFlow, Keras, OpenCV

Software Engineer Intern Jan. 2021 – Jun. 2021
Kairos Sensing *Sri Lanka*

- Developed Sri Lanka's first IMU sensor-based full-body 3D motion capture system (github)
- Implemented core motion processing algorithms and real-time visualization in Java and C++
- Integrated sensor data pipelines with hardware and firmware teams
- Tech Stack: Java, C++, jMonkeyEngine

Education

University of Moratuwa Sri Lanka
B.Sc.(Hons) Biomedical Engineering *Jan. 2020 – Jun. 2024*

- Cumulative GPA: 4.04/4.20 | First Class (Honours)

Royal College, Colombo - 07 Sri Lanka
G.C.E. Advanced Level Examination (Physical Science) *Jun. 2016 - Aug. 2018*

- Z-Score: 2.4 (3As) | Ranked 83rd in the country

Publications

Real-Time AI-Driven People Tracking and Counting Using Overhead Cameras | IEEE-Explore 2024
Presented in TENCON 2024 - 2024 IEEE Region 10 Conference (TENCON) - Singapore

Personal & Academic Projects

Gargantua | github 2026

Open-source operator-grade control plane for MCP-powered LLM agents

- Built a platform to define agents, teams, and MCP servers as database rows with full CRUD, RBAC, and audit logging
- Implemented AES-256-GCM envelope encryption with zero-downtime KEK rotation and RS256 JWT authentication
- Designed a warm-handle MCP connection cache with ref-counting, leasing, and idle reaping
- Developed a Next.js admin and chat console with SSE streaming, packaged in a single self-hosted container
- Tech Stack: Python, FastAPI, Agno, PostgreSQL, Next.js, TypeScript, Docker

Computational Microscope Design and 3D Image Reconstruction | github 2024

Undergraduate thesis in collaboration with Harvard University

- Investigated deep learning approaches for solving ill-posed inverse problems in 3D image reconstruction
- Designed and modeled optics-inspired computational imaging systems
- Tech Stack: Python, MATLAB, PyTorch

SINDiB – Micromouse | github 2023

Micromouse robot developed for RoboFest 2023

- Integrated a flood-fill maze-solving algorithm on an STM32 microcontroller
- Configured motors and interfaced an OLED display via SPI communication
- Tech Stack: C, C++, STM32CubeIDE

Skills

Programming Languages: Python, Java, Go, SQL, TypeScript, Dart, MATLAB,
Frameworks & Libraries: Django, Spring Boot, FastAPI, Cerbos, Angular, React, Flutter, Pipecat, WebRTC, Liquibase
Databases: PostgreSQL, Oracle, MongoDB, Neo4j, Redis
DevOps & Tools: Kubernetes, ArgoCD, Jenkins, Docker, Git, GitHub
ML / AI / CV: PyTorch, TensorFlow, Keras, OpenCV

References

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Tishan Harischandra – Associate Architect, Cloud Solutions International, Sri Lanka.
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